

Rotational dynamics

SS ILP Physics (Intermediate), Week 6 Class 1



Outline

- Newton's Second Law for Rotation (Torque law)
- Static equilibrium
- Rotational and translational quantities
- Rotational kinetic energy

Newton's Second Law for Rotation

- Torque about an axis of rotation produces angular acceleration about the axis

$$\sum \vec{\tau} = I_s \vec{\alpha}$$

I_s is the moment of inertia about the axis of rotation

$\vec{\alpha}$ is the angular acceleration about the axis of rotation

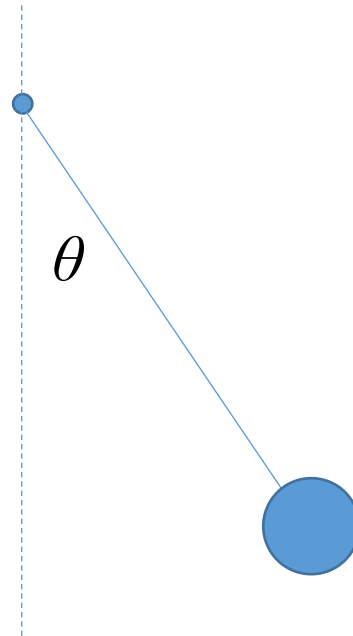
- Analogous to Newton's Second Law for Linear Motion

$$\sum \vec{F} = m \vec{a}$$

http://physics.bu.edu/~redner/211-sp06/class13/class14_newton2.html

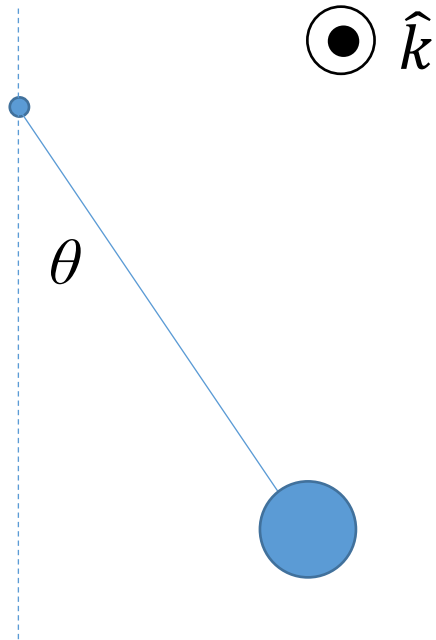
Example

A small ball with mass m is attached to one end of a massless string with length of l , as shown. When the ball is θ from the vertical, find the net torque exerted on the ball about the pivot.



Example - Solution

A small ball with mass m is attached to one end of a massless string with length of l , as shown. When the ball is θ from the vertical, find the net torque exerted on the ball about the pivot.



$$\tau_T = Tl \sin 180^\circ = 0$$

$$\tau_G = mgl \sin \theta$$

$$\vec{\tau} = \vec{\tau}_T + \vec{\tau}_G = mgl \sin \theta (-\hat{k})$$

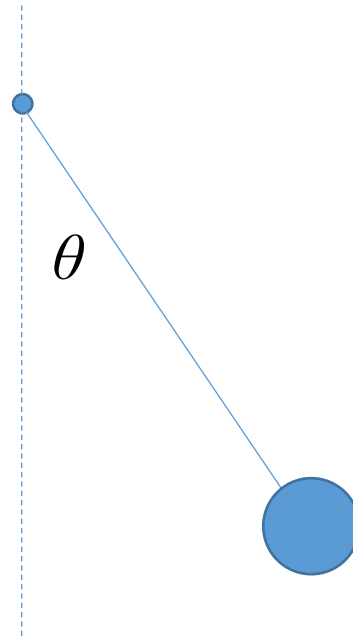
Solving rotational dynamics problems

- I. Identify the forces acting on the object of interest.
- II. Draw free body diagram of the object.
- III. Choose an appropriate coordinate system.
- IV. Identify the pivot point and find the torques about the pivot due to different forces.
- V. According to the shape of the object and pivot point, find the moment of inertia.
- VI. Apply torque law $\sum \vec{\tau} = I_s \vec{\alpha}$
- VII. Consider constraints if any for solving unknowns.

<https://openstax.org/books/university-physics-volume-1/pages/10-7-newtons-second-law-for-rotation>

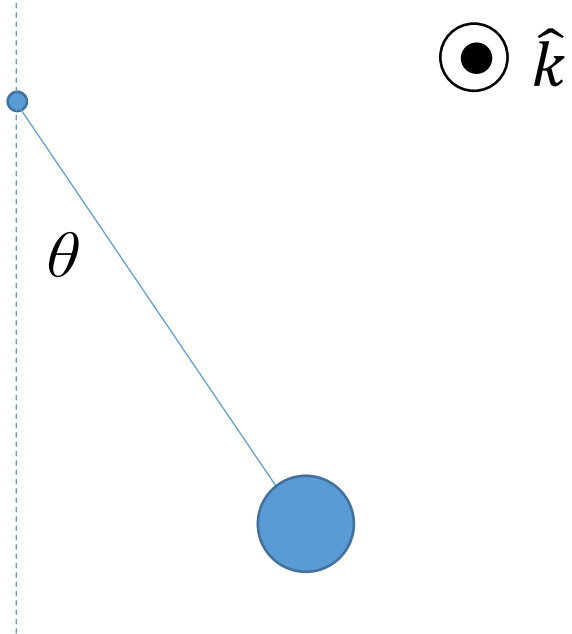
Example

A small ball with mass m is attached to one end of a massless string with length of l , as shown. When the ball is θ from the vertical, find the acceleration experienced by the ball about the pivot.



Example - Solution

A small ball with mass m is attached to one end of a massless string with length of l , as shown. When the ball is θ from the vertical, find the angular acceleration experienced by the ball about the pivot.



$\odot \hat{k}$

$$\tau_T = Tl \sin 180^\circ = 0$$

$$I_s = ml^2$$

$$\tau_G = mgl \sin \theta$$

$$\vec{\tau} = \vec{\tau}_T + \vec{\tau}_G = -mgl \sin \theta \hat{k}$$

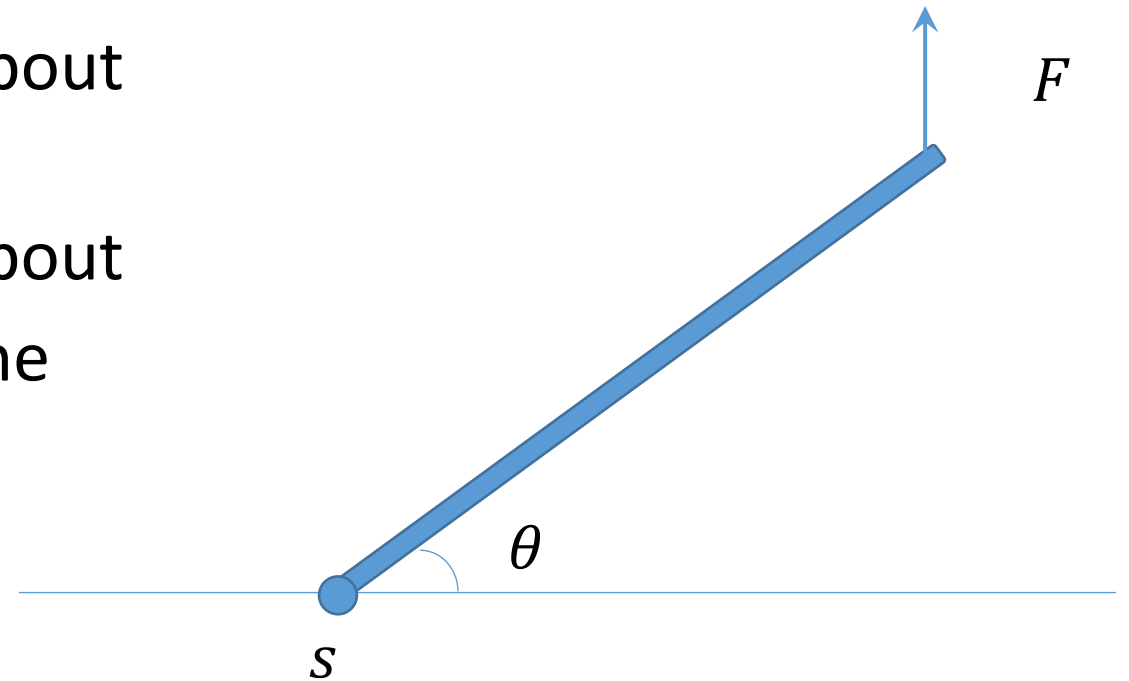
$$\vec{\tau} = I_s \vec{\alpha} \longrightarrow -mgl \sin \theta = I_s \alpha$$

$$\vec{\alpha} = -\frac{mgl \sin \theta}{ml^2} \hat{k}$$

Concept Question

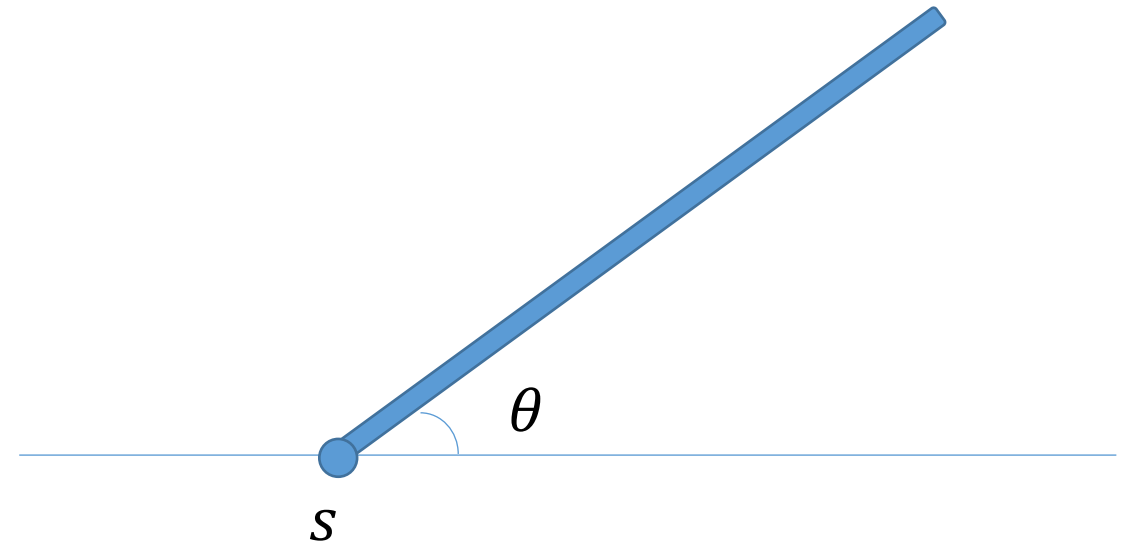
A force of magnitude F is exerted on one end of a rod of length l , as shown. Which of the following statements is true?

1. The net torque exerted on the rod about point s is due to $\vec{F}$ and the gravity.
2. The net torque exerted on the rod about point s is due to $\vec{F}$, the gravity and the force applied by the hinge.
3. No idea.



Question

A rod of length l and mass m is connected to a hinge. Find the angular acceleration of the rod when it is at an angle of θ above the horizontal. The moment of inertia of the rod about the hinge is $\frac{1}{3}ml^2$.



Static equilibrium

- Translation equilibrium $\sum_i \vec{F}_i = 0$

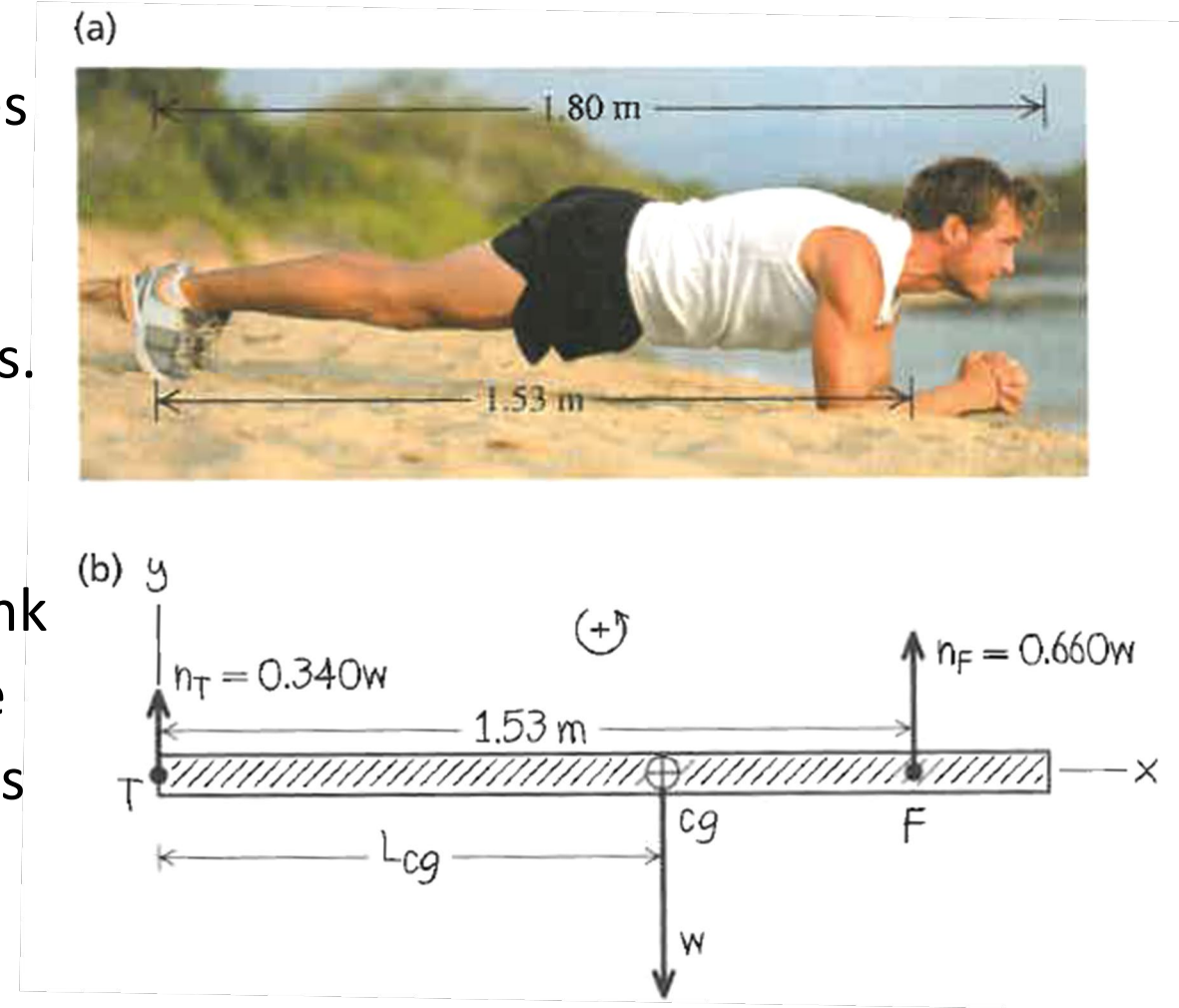
- Rotational equilibrium $\sum_i \vec{\tau}_i = 0$

- Static equilibrium $\sum_i \vec{F}_i = 0$ and $\sum_i \vec{\tau}_i = 0$

<https://openstax.org/books/university-physics-volume-1/pages/12-1-conditions-for-static-equilibrium>

Example

Holding plank position with a scale under his toes and another under his forearms, one athlete measured that 0.66 of his weight was supported by his forearms and 0.34 of his weight by his toes. That is the total normal forces on his forearms and toes were $0.66w$ and $0.34w$, respectively, where w is his weight. He is 1.8m tall, and in plank position the distance from his toes to the middle of his forearms is 1.53m. How far from his toes is his center of gravity?



H. D. Young and R. A. Freedman, University Physics (14th edition), Example 11.2, p368

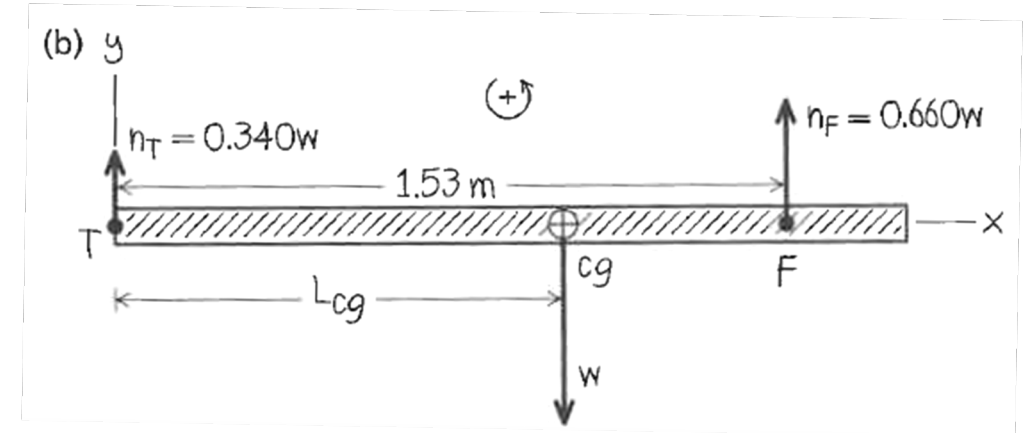
Example - Solution

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$$\sum \vec{F}_y = 0 \rightarrow n_T + n_F - mg = 0$$

$$\sum \vec{\tau} = 0 \rightarrow n_T L_{cg} - n_F \times (1.53 - L_{cg}) = 0$$

$$L_{cg} = \frac{1.53 n_F}{n_T + n_F} = 1.01 \text{ m}$$

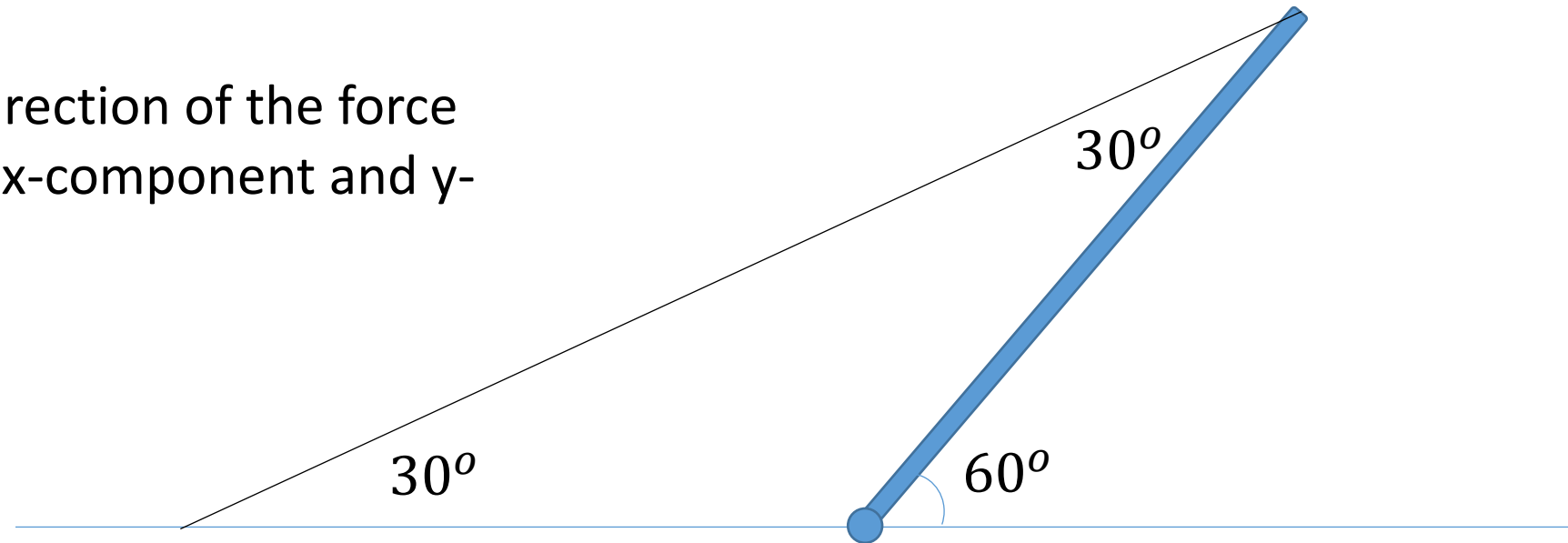


H. D. Young and R. A. Freedman, University Physics (14th edition), Example 11.2, p368

Example

A rod is of length l and mass m is in static equilibrium. One end of the rod is connected to the surface via a hinge and the other end of the rod is connected to the surface via a string. The length of the string is L . Find the tension due to the string and the force due to the hinge.

Hint: without knowing the direction of the force due to the hinge, we find its x-component and y-component separately.



<https://www2.tntech.edu/leap/murdock/books/v2chap3.pdf>

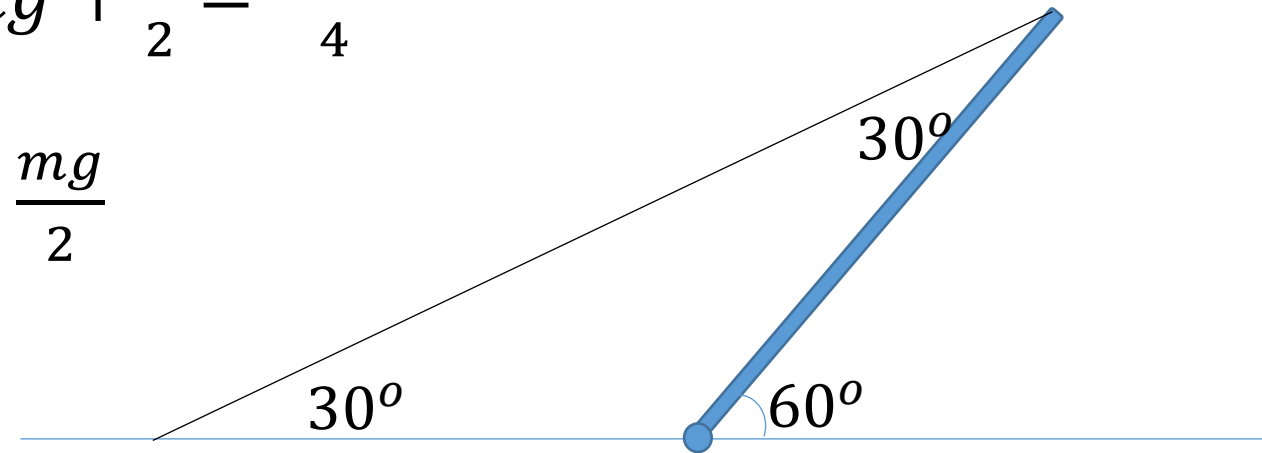
Example - Solution

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$$F_{h,x} - T \sin 60^\circ = 0 \rightarrow F_{h,x} = \frac{\sqrt{3}T}{2} = \frac{\sqrt{3}}{2} \frac{mg}{2}$$

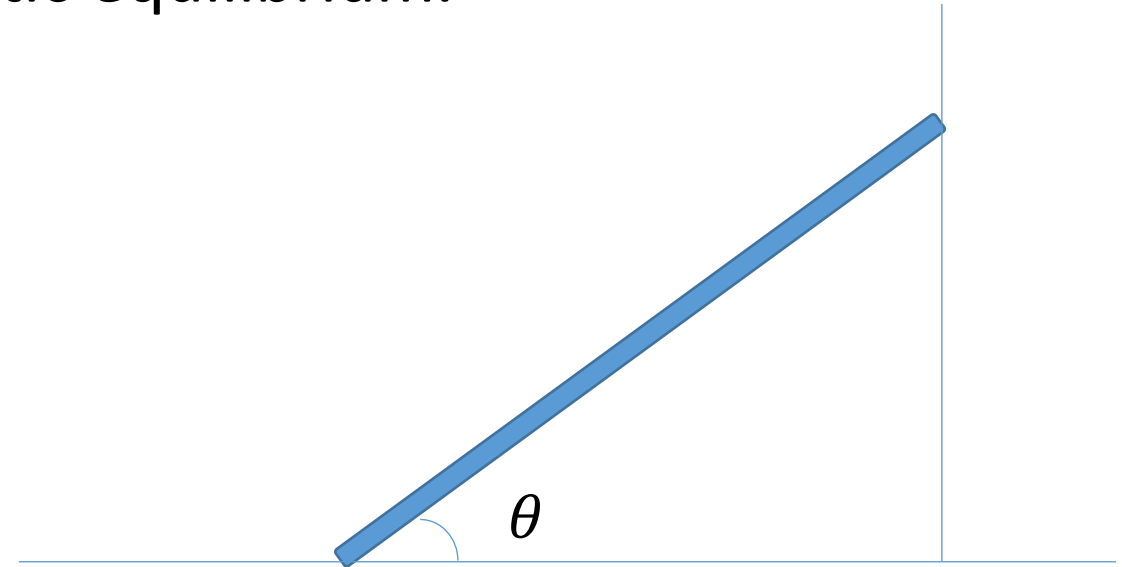
$$F_{h,y} - mg - T \cos 60^\circ = 0 \rightarrow F_{h,y} = mg + \frac{T}{2} = \frac{5mg}{4}$$

$$Tl \sin 150^\circ - \frac{mg}{2} l \sin 150^\circ = 0 \rightarrow T = \frac{mg}{2}$$



Question

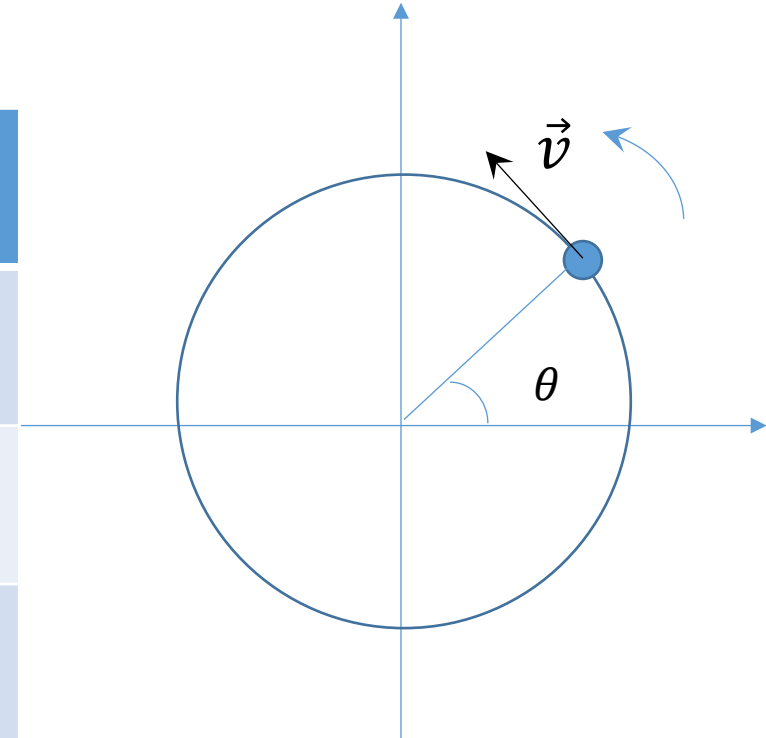
A rod is of length l and mass m , leaning against the frictionless vertical wall. The ground is not frictionless, but with a static friction coefficient μ_s . Find the range of θ in terms of μ_s to ensure the rod is in static equilibrium.



[https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_I_-_Classical_Mechanics_\(Gea-Banacloche\)/09%3A_Rotational_Dynamics/9.05%3A_Statics](https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_I_-_Classical_Mechanics_(Gea-Banacloche)/09%3A_Rotational_Dynamics/9.05%3A_Statics)

Rotational and translational quantities

Rotational quantity	Translational quantity	Relation
θ	s	$s = r\theta$
ω	v	$v = r\omega$
α	a_t	$a_t = r\alpha$

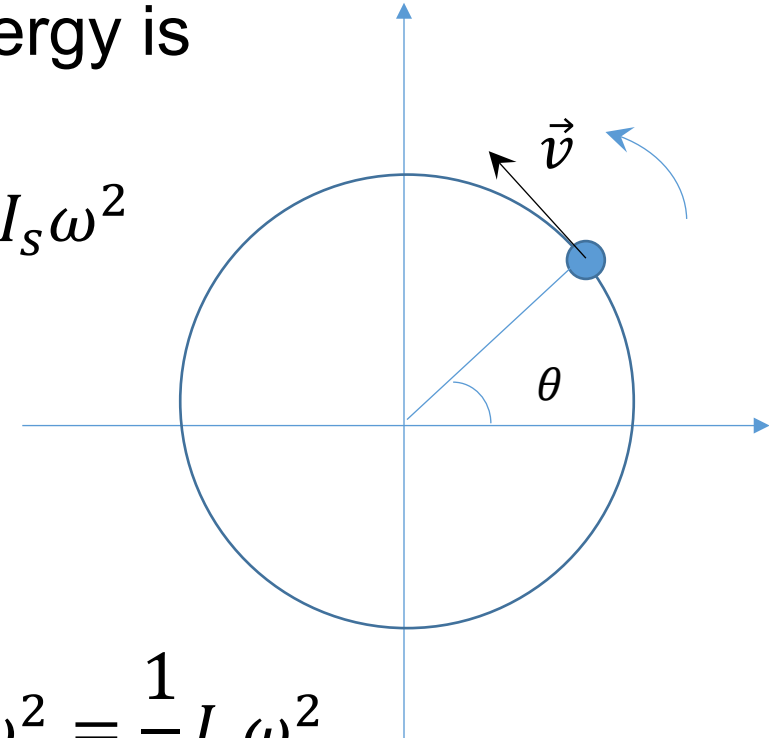


<https://openstax.org/books/university-physics-volume-1/pages/10-3-relating-angular-and-translational-quantities>

Rotational kinetic energy

- For a point particle undergoing rotation, its kinetic energy is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2 = \frac{1}{2}mr^2\omega^2 = \frac{1}{2}I_s\omega^2$$



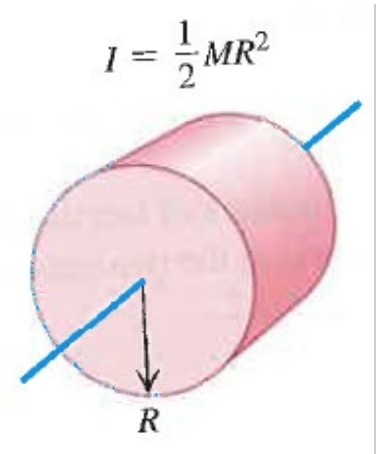
- For a system of multiple objects rotating with ω

$$K = \frac{1}{2}\sum_i m_i v_i^2 = \frac{1}{2}\sum_i m_i (r_i \omega)^2 = \frac{1}{2}\sum_i m_i r_i^2 \omega^2 = \frac{1}{2}I_s \omega^2$$

<https://openstax.org/books/university-physics-volume-1/pages/10-4-moment-of-inertia-and-rotational-kinetic-energy>

Example

A 6.0-kg cylinder with radius of 0.1 m is rotating about the axis through its center with angular speed $\omega = 5.0$ rad/s. The moment of inertia of a cylinder rotating about the axis through its center is $I_s = \frac{1}{2}MR^2$. Find its kinetic energy.



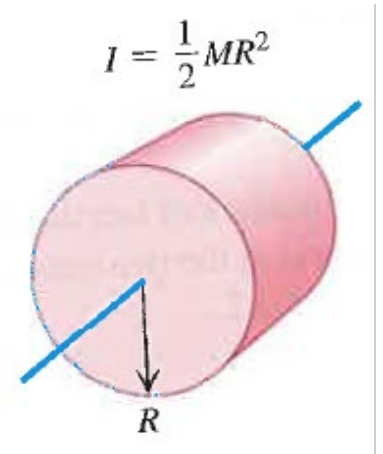
H. D. Young and R. A. Freedman, University Physics (14th edition), p310

Example: Solution

A 6.0-kg cylinder with radius of 0.1 m is rotating about the axis through its center with angular speed $\omega = 5.0$ rad/s. The moment of inertia of a cylinder rotating about the axis through its center is $I_s = \frac{1}{2}MR^2$. Find its kinetic energy.

$$I_s = \frac{1}{2}MR^2 = \frac{1}{2} \times 6 \times (0.1)^2 = 0.03 \text{ kg} \cdot \text{m}^2$$

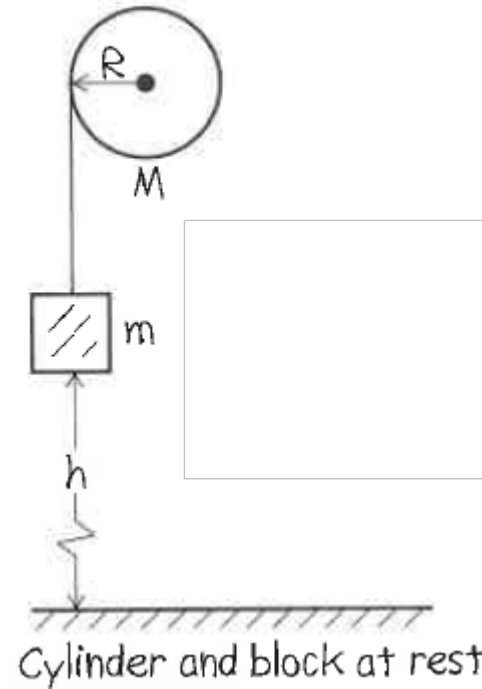
$$K = \frac{1}{2}I_s\omega^2 = 0.375 \text{ J}$$



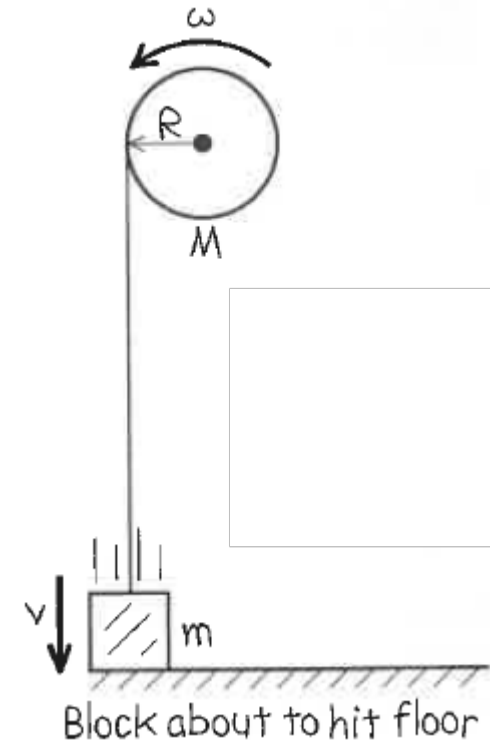
Question

We wrap a light, non-stretching cable around a solid cylinder with mass M and radius R . The cylinder rotates with negligible friction about a stationary horizontal axis. We tie the free end of the cable to a block of mass m and release the block from rest at a distance h above the floor. As the block falls, the cable unwinds without stretching or slipping. Find the speed of the falling block and the angular speed of the cylinder as the block strikes the floor. The moment of inertia of a cylinder is $I_S = \frac{1}{2}MR^2$.

(a) Initial (block at point 1)



(b) Final (block at point 2)



H. D. Young and R. A. Freedman, University Physics (14th edition), Example 9.8, p311